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Effect of combined pretreatment of waste activated sludge for anaerobic digestion process

Hui Yi, Yun Han*, Yang Zhuo

School of Municipal and Environmental Engineering, Xi'an University of Architecture and Technology, Xi'an 710055, China

Abstract

Alkaline and low-temperature thermal pretreatment has been used to the subsequent anaerobic digestion of waste activated sludge (WAS) respectively. Different combinations of these two methods were investigated and biochemical methane potential (BMP) test was used to assess the anaerobic digestibility of pretreated WAS. In this study, an optimal reaction condition was obtained by combined pretreatment and BMP test. The combination was after alkalized for 24h, adding 0.05 g NaOH /g TS with 9 hours at 70℃. Under this condition the removal of SS was achieved 21% and soluble chemical oxygen demand (SCOD) was more than 200 times of control group. The ratio of soluble carbohydrate/total carbohydrate can reach 72.8%. For BMP test, it's nearly 6 times higher biogas production was obtained than the control and the average value of methane content of biogas production is 64 %. Thus, combined (alkaline + low-temperature thermal) pretreatment was efficiency for solubilization and biogas production.

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Keywords: Anaerobic digestion; Pretreatment; Alkaline;Low-temperature thermal pretreatment; BMP

1. Introduction

The considerable increase in the amount of sewage sludge production in wastewater treatment plant (WWTP) becomes a serious and pressing problem to the rapidly and healthy development of the city [1]. Before disposing the sewage sludge, it has to be stabilized sufficiently to reduce its organic content, odor problem and pathogen contamination. Neyens reported on that sludge handling and disposal already accounts for up to 50% of total treatment costs of wastewater purification in 2004[2].

Anaerobic digestion is common method for sludge stabilization to achieve sludge reduction, harmlessness and resource reuse. Compared with other processes, its advantages are less energy required, a better stabilized product and useable gas [3]. Anaerobic digestion has three basically steps: hydrolysis,

* Corresponding author. Tel.: 013991207226;.
E-mail address: hanyun@xauat.edu.cn

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common organic substance as well as hydrogen, oxygen and nitrogen [13]. After the treatment, the max value can reach 11495 mg/L, it is more than 90 times of the control sludge (125.5 mg/L). That is to say, the NaOH dose adding is a major factor which influenced the protein release.

From all above, we know that the alkaline treatment can resale more soluble organic materials when combined with thermal pretreatment. Furthermore, from Table 2 we found that thermal pretreatment is more effective for low dose NaOH than high dose NaOH.

3.2. BMP test

For assessing the efficiency of A+T pretreatment anaerobic digestibility and evaluating the biogas production, BMP test was used. The sample sludge were treated by adding 0.05 g NaOH /g TS (A24h+T9h) at various temperature (50℃, 60℃, 70℃, 80℃, 90℃), the NaOH dose and thermal treatment holding time were obtained from the combined pretreatment (A+T) in this work. Biogas production and the reduction of TCOD, VSS can illustrate the efficiency of anaerobic biological degradation. SCOD, VFA, soluble carbohydrate and protein can be readily used to produce methane during anaerobic digestion [14].

Fig.1 shows the cumulative biogas production of BMP test. The daily biogas productions increase quickly at fist 3 days, then decrease rapidly in 4 days, after 10 days the daily biogas production tend to flat keep in a low amount. Ray had pointed that pretreatment with sodium hydroxide increased gas production by between 29% and 112% over the control sludge[15].The accumulated biogas production at the end of 30 days of the digestion period is nearly 45 mL for control sample, and at 70℃ (A24h+T9h) it is around 329 mL, nearly 630% higher biogas production is obtained than the control sludge，it is much more efficiency than untreated sludge. The average value of methane content of biogas production is 64 %. Fig.2 shows the data about VSS of BMP test. VSS contents 95% organic matters in WAS. The reduction of VSS can indicate the organic matters removal. Similarly, it gets the max value of VSS reduction (27.1%) at 70℃(A24h+T9h), this is similar to the result of Rani [9] .

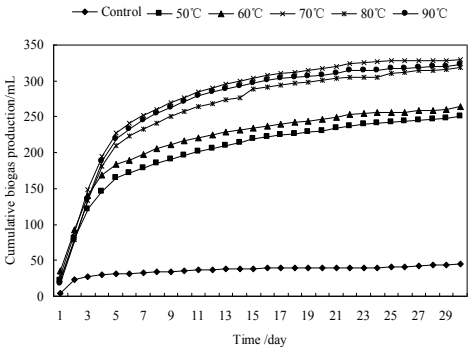

Fig. 1. Cumulative biogas production of BMP test .

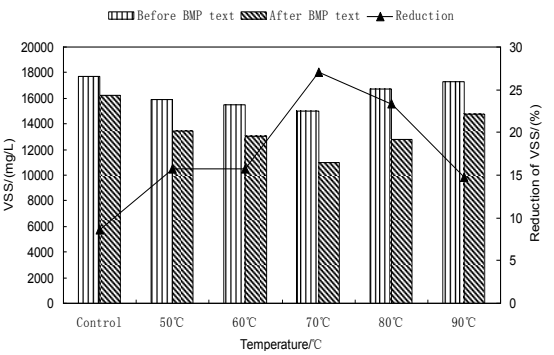

Fig. 2. VSS of BMP test.

4. Conclusions

The results of this study indicated that the alkaline + low-temperature thermal pretreatment has the potential to damage excess sludge structure and cell membranes and to release extracellular and possibly intracellular compounds with high solubility. It also increased the bioavailability of waste activated

sludge components under thermophilic BMP tests. Under the conditions of 70℃, adding 0.05 g NaOH /g TS with 9 hours thermal holding time, SS reduction achieved 21%, SCOD value was nearly 16 times of the sludge treated A24h+T2h and more than 200 times of control group. For BMP test, it's nearly 83% higher biogas production was obtained than the raw sludge and got 27.1% VSS reduction. So the solubilization and anaerobic digestibility for the treatment of alkaline + low-temperature is satisfactory.

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